Vasukumar_Patel_Lab5.R | data

Source on Save | Run

```r
library(readxl)
data <- read_excel("H:/My Drive/Level_4/R language/Lab5/Ascent_STS121_Data.xlsx")

print(names(data))

colnames(data)[1:2] <- c("time", "altitude")

time <- data$time
altitude <- data$altitude

png("H:/My Drive/Level_4/R language/Lab5/time_vs_altitude.png")

plot(time, altitude,
     main = "Time vs Altitude",
     xlab = "Time",
     ylab = "Altitude",
     pch = 16)

dev.off()


#Functions

# Forward Divided Difference
student.numDiff.fdd <- function(time, altitude) {

  n <- length(time)
  result <- rep(NA, n)

  for (i in 1:(n-1)) {
    result[i] <- (altitude[i+1] - altitude[i]) / (time[i+1] - time[i])
  }

  return(result)

}
```

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```r
# Backward Divided Difference
student.numDiff.bdd <- function(time, altitude) {
  n <- length(time)
  result <- rep(NA, n)

  for (i in 2:n) {
    result[i] <- (altitude[i] - altitude[i-1]) / (time[i] - time[i-1])
  }

  return(result)
}

# Central Divided Difference
student.numDiff.cdd <- function(time, altitude) {
  n <- length(time)
  result <- rep(NA, n)

  for (i in 2:(n-1)) {
    result[i] <- (altitude[i+1] - altitude[i-1]) / (time[i+1] - time[i-1])
  }

  return(result)
}

# Second Derivative
student.numDiff.secDer <- function(time, altitude) {
  n <- length(time)
  result <- rep(NA, n)

  for (i in 2:(n-1)) {
    h <- time[i] - time[i-1]
    result[i] <- (altitude[i+1] - 2*altitude[i] + altitude[i-1]) / (h^2)
  }

  return(result)
}
```

```r

#Calculations
fdd <- student.numDiff.fdd(time, altitude)
bdd <- student.numDiff.bdd(time, altitude)
cdd <- student.numDiff.cdd(time, altitude)
acc <- student.numDiff.secDer(time, altitude)

png("H:/My Drive/Level_4/R language/Lab5/velocity_plot.png")

plot(time, fdd,
     col = "red",
     pch = 16,
     main = "Velocity vs Time",
     sub = "FDD (Red Circle), BDD (Blue Triangle), CDD (Green Cross)",
     xlab = "Time",
     ylab = "Velocity")

points(time, bdd, col = "blue", pch = 17)
points(time, cdd, col = "green", pch = 3)

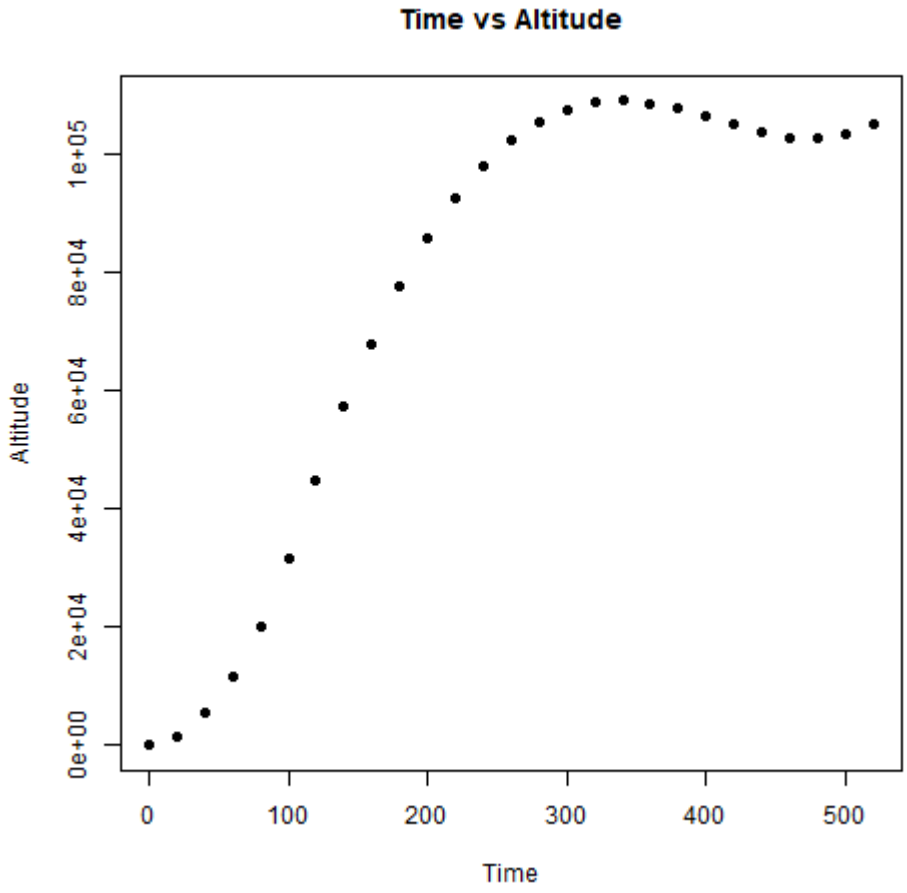
dev.off()

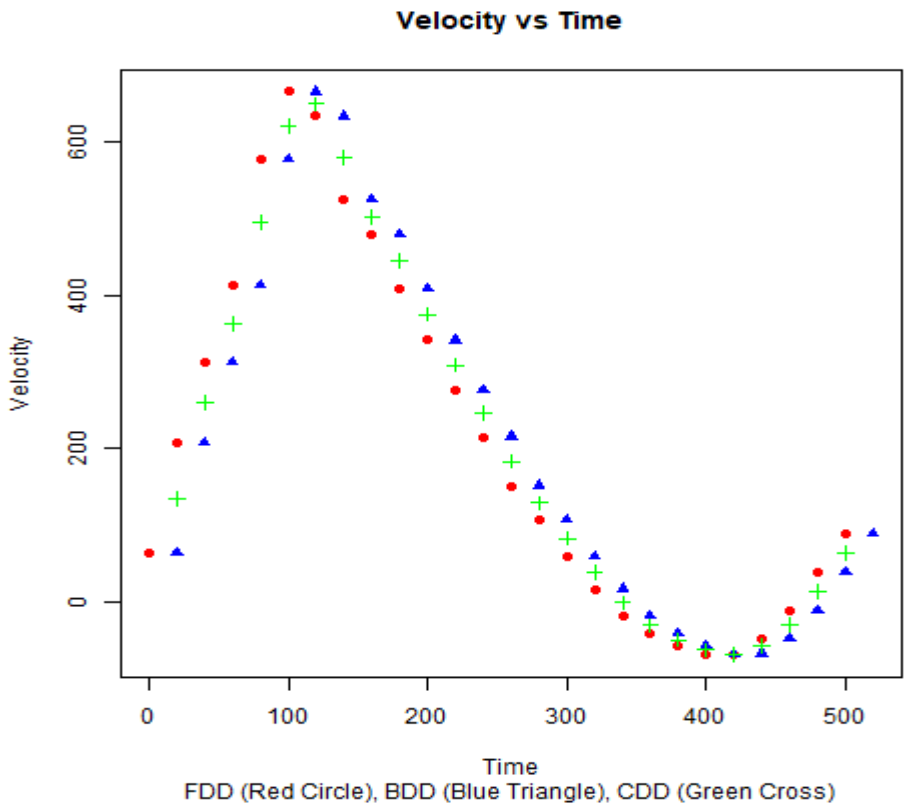
png("H:/My Drive/Level_4/R language/Lab5/acceleration_plot.png")

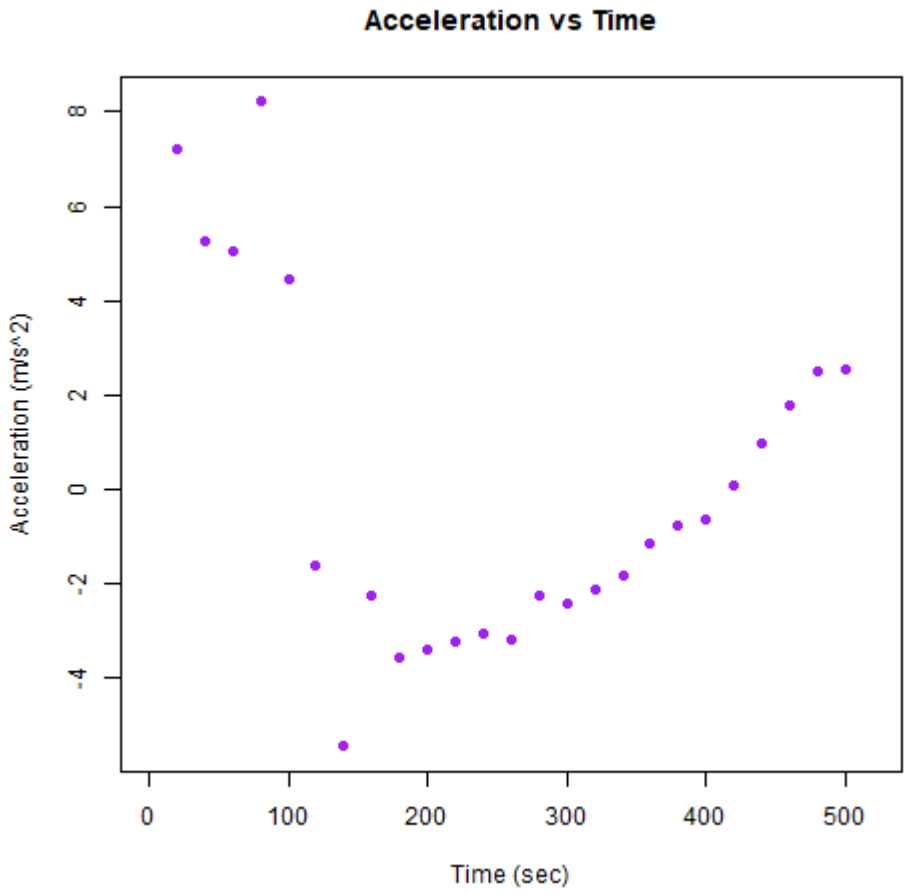
plot(time, acc,
     col = "purple",
     pch = 16,
     main = "Acceleration vs Time",
     xlab = "Time (sec)",
     ylab = "Acceleration (m/s^2)")

dev.off()

```




Velocity vs Time
Velocity
0
200
400
600
0
100
200
300
400
500
Time
FDD (Red Circle), BDD (Blue Triangle), CDD (Green Cross)


Acceleration vs Time
Acceleration (m/s^2)
-4
-2
0
2
4
6
8
0
100
200
300
400
500
Time (sec)

1. Why is there only 1 data point at 0 sec?

➔ At 0 seconds, there is only one data point because we cannot look at a value before 0.
The backward and central methods need a previous value, but it does not exist.
Only the forward method works, so only one point appears.

2. Why is there only 1 data point at 520 sec?

➔ At 520 seconds, there is only one data point because we cannot look at a value after 520.
The forward and central methods need a next value, but it does not exist.
Only the backward method works, so only one point appears.